

Frailty as a Modifier of the Associations Between Vascular Risk Factors and Incident Dementia

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Abstract

Background and Objectives

The contribution of late-life vascular risk factors to dementia risk remains controversial. Because low blood pressure (BP) has been associated with worse clinical outcomes in frail individuals, we hypothesized hypertension, but not diabetes or smoking, is associated with higher dementia risk in robust than in frail older adults.

Methods

We performed a prospective cohort analysis of the Atherosclerosis Risk in Communities Neurocognitive Study (ARIC-NCS) over 11 years (2011–2022). We included all community-living White and Black participants aged 67–89 years without dementia at baseline visit 5 (2011–2013) from ARIC-NCS field centers (Jackson, Mississippi; Forsyth County, North Carolina; Washington County, Maryland; Minneapolis suburbs, Minnesota). The primary vascular risk factors measured at baseline included elevated BP (systolic BP 120–129 mm Hg and diastolic BP < 80 mm Hg), hypertension (systolic BP ≥ 130 mm Hg, diastolic BP ≥ 80 mm Hg, or use of medication for BP), diabetes (fasting glucose ≥ 126 mg/dL, nonfasting glucose ≥ 200 mg/dL, self-reported physician's diagnosis, or use of diabetes medication), and former and current smoking (self-reported). We defined frailty status using the Fried criteria (5 components that include low energy, low physical activity, slowness, weakness, and weight loss). We estimated cause-specific hazard ratios (HRs) of incident dementia (ascertained from in-person neuropsychological assessments, semiannual participant or informant report, or surveillance of claims from hospitalizations and death certificates) with Cox proportional hazards models including a multiplicative interaction between vascular factors and frailty. HRs of dementia were then stratified by frailty (robust [no frailty components present] vs prefrail/frail [at least 1 frailty component present]).

Results

There were 377 (15.8%) dementia cases in robust participants ($n = 2,383$; mean age, 74.2 years; 55.4% female) and 812 (30.0%) in prefrail/frail participants ($n = 2,710$; mean age 76.4 years; 61.3% female). There was a significant interaction between BP and frailty on dementia risk ($p = 0.026$). For robust participants, HRs were 1.03 (95% CI 0.65–1.64) for elevated BP and 1.39 (95% CI 1.00–1.94) for hypertension relative to normal BP. For prefrail/frail participants, HRs were 0.68 (95% CI 0.49–0.95) and 0.82 (95% CI 0.66–1.01), respectively. Diabetes and current smoking were associated with higher dementia risk in both robust and prefrail/frail individuals.

Discussion

Late-life hypertension was associated with a lower relative risk of dementia in prefrail/frail participants, but the association was positive in robust participants. BP interpretation and management to support brain health in older adults could consider age-related functional status.

MORE ONLINE

Supplementary Material

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Glossary

APOE = apolipoprotein; **ARIC-NCS** = Atherosclerosis Risk in Communities Neurocognitive Study; **BMI** = body mass index; **BP** = blood pressure; **DBP** = diastolic blood pressure; **HR** = hazard ratio; **SBP** = systolic blood pressure.

Introduction

Treating vascular risk factors (hypertension, diabetes, and smoking) is increasingly recognized as foundational to maintain brain health and reduce risk of dementia, in addition to its established role in cardiovascular disease prevention.^{1,2} Hypertension, diabetes, and smoking in midlife are prevalent³ and principal drivers of dementia risk.^{4,5} The dominant mechanism is presumed to be cerebral small vessel disease that contributes to cognitive decline⁶; however, vascular risk factors could also indirectly modify Alzheimer disease neuropathology.^{5,7} Their stronger associations with dementia incidence,⁸ as well as closer proximity to dementia expression relative to other internationally recognized risk factors (e.g., physical inactivity and obesity), also supports causal associations. Midlife vascular risk factor control might therefore considerably reduce dementia risk by delaying dementia caused by either process.⁹

The contribution of vascular risk factors present in late life to dementia is controversial, however, and there is much less certainty for hypertension. Whereas late-life diabetes and smoking have inconsistently been linked with elevated dementia risk,⁴ studies have reported null and inverse associations between late-life hypertension and dementia.^{10,11} This raises concerns for clinical decisions regarding blood pressure (BP) treatment. There are several proposed explanations for the association of lower BP in later life (systolic blood pressure [SBP] <120 mm Hg, diastolic blood pressure [DBP] <80 mm Hg) with poor cognitive outcomes. It is hypothesized that higher BP levels than the target range of an intensive BP treatment strategy (e.g., SBP <120 mm Hg) might be needed to perfuse the older brain, and prevention of hypoperfusion could be particularly important after chronic midlife hypertension with stiffening of blood vessels.^{10,11} Alternatively, noncausal mechanisms may be explanatory. Age-related comorbidities such as heart failure or heart disease might be strongly associated with lower BP in late life, which also in turn increase risk of dementia.^{10,11}

Frailty status could provide context for the observed conflicting research for hypertension and act as a moderator of the association between late-life BP and dementia.^{12,13} Although there is no consensus on a definition, frailty can be defined as an age-related clinical state marked by decline across many physiologic systems—including metabolic, endocrine, and immune responses—and a decreased response to stressors.^{14,15} Within the physical frailty phenotype framework, there are 5 components of low energy, low physical activity, slowness, weakness, and weight loss.¹⁶ The presence of at least 3 components is indicative of frailty, 1 or 2 as prefrail, and none as robust. It is of

importance that low BP specifically has been associated with poorer clinical outcomes in frail older adults and high BP with worse outcomes in robust individuals.^{12,13} These findings are hypothesized to be attributable to an increase in vascular stiffness, which can be a result of greater pulse pressure or changes in the artery wall, in frail individuals.¹³ Frailty status might therefore characterize individuals who are more vulnerable to having low BP. Given prior studies documented inverse associations of late-life hypertension with lower dementia risk, but not for diabetes or smoking, frailty may not modify dementia risk associated with diabetes or smoking as it does for hypertension.

The Atherosclerosis Risk in Communities Neurocognitive Study (ARIC-NCS) was initiated to investigate vascular contributions to dementia in community-living older adults.¹⁷ Previous ARIC-NCS studies reported weak or inverse associations between late-life hypertension,^{18,19} and weak associations for diabetes,²⁰ with incident dementia. In this study, we expanded upon prior work to evaluate whether frailty status, defined using physical frailty criteria, modifies associations of vascular risk factors with incident dementia. This analysis also includes longer follow-up with a greater number of dementia cases than prior investigations. We hypothesized that late-life (baseline age 67–89 years) hypertension, but not diabetes or smoking, is associated with an elevated risk of dementia in robust (no frailty components present) adults but not in prefrail/frail (at least 1 frailty component present) adults.

Method

Study Design and Sample

ARIC is a prospective cohort study that recruited 15,792 participants aged 45–64 years in 1987–1989 from 4 US communities (the city of Jackson, Mississippi; Forsyth County, North Carolina; selected suburbs of Minneapolis, Minnesota; Washington County, Maryland).¹⁷ We used risk factor data collected at in-person visit 5 (2011–2013), which we set as baseline for this analysis given the age of participants. Among 6,538 participants who attended visit 5, we included 6,135 participants that self-identified as White and Black race in line with ARIC recommendations, with complete follow-up, and without baseline dementia ($n = 403$ excluded; eFigure 1). In sensitivity analyses, we repeated the primary analysis in the complete-case sample ($n = 5,093$; eFigure 1).

Incident Dementia

We identified all-cause dementia cases using a standardized ARIC algorithm.^{8,21,22} Diagnosis incorporates individual and proxy-reported cognitive, behavioral, and functional data from ARIC visits as well as from annual and semiannual

follow-up calls^{23,24} and community surveillance of International Classification of Disease (ninth and tenth edition) claims data from hospitalizations and death certificates.^{25,26} The procedures detect dementia cases in participants who do not attend, which mitigates drop-out and selection bias. Many cases were identified in participants who did not attend in-person assessments, which would underestimate dementia incidence and decrease precision of relative risk estimates had they not been identified.²¹ Algorithmic diagnoses are confirmed by an expert panel of physicians and neuropsychologists blinded to risk factor status in accordance with Diagnostic and Statistical Manual of Mental Disorders (fifth edition) criteria.²⁷

Death was ascertained through ARIC's surveillance methods. This includes contact of participant proxies and linked data from vital statistics from the National Death Index, death certificates, hospital records, and obituaries.

Vascular Risk Factors

We defined hypertension (3 BP measurements obtained, averaging the last 2) in accordance with current clinical criteria²⁸: elevated BP, SBP (in mm Hg) 120–129 and DBP (in mm Hg) < 80; hypertension, SBP ≥ 130 and DBP ≥ 80; or use of medication for BP. In separate models, we delineated stage 1 hypertension as SBP 130–139 and DBP 80–89 and stage 2 hypertension as SBP ≥ 140 or DBP ≥ 90. In sensitivity analyses, we used prior JNC-7 criteria to define hypertension²⁹: prehypertension, SBP 120–139 or DBP 80–89; hypertension, SBP ≥ 140, DBP ≥ 90; or use of BP medication. We defined diabetes as a fasting glucose level ≥ 126 mg/dL, nonfasting glucose level ≥ 200 mg/dL, self-reported physician's diagnosis of diabetes, or use of medication for diabetes. We defined former and current smoking status by self-report.

Frailty Status

We used a composite variable previously validated in ARIC³⁰ based on the Fried physical (phenotypic) frailty criteria.¹⁶ In ARIC, frailty status incorporates 5 measures that includes exhaustion, weight loss, low energy expenditure, slow walking speed, and low grip strength. Exhaustion was defined by answering "some of the time" or "most of the time" to either question of *I felt everything I did was an effort* or *I could not get "going"* on the Center for Epidemiologic Studies-Depression Scale.³¹ Weight loss was defined as either a 10% reduction in weight from visit 4 to visit 5 or a measured body mass index (BMI) ≤ 18.5 at baseline. Low energy expenditure was defined as the bottom 20th percentile (sex specific) of the frailty validation cohort on the Baecke Leisure Sports Activity Index.³² Slow walking speed was defined from participants' usual 4-meter walking speed as the bottom 20th percentile (sex and height adjusted) of the frailty validation cohort.¹⁶ Low grip strength was defined as the lowest 20th percentile (sex and BMI specific) of the frailty validation cohort¹⁶ in grip strength measured in the preferred hand with a hydraulic handgrip dynamometer.

Robust status was defined as no frailty components present, prefrail as 1–2 components present, and frail as ≥ 3 components present.¹⁶ We dichotomized the composite variable (robust [reference level], prefrail/frail) for the primary analysis because of sample size considerations. In additional analyses, we used the 3-level variable (robust [reference level], prefrail, and frail).

Covariates

Variables obtained at study entry were birthdate (used to calculate age in years), sex (male or female), race (White or Black), educational attainment (lower than completed high school, high school or equivalent, or at least some college), and apolipoprotein (APOE) ε4 genotype (0 or ≥ 1 ε4 allele) collected using the TaqMan assay (Applied Biosystems, Foster City, CA). Other variables collected at visit 5 included self-reported alcohol consumption (never, former, or current) and plasma total and high-density cholesterol (mg/dL). Sensitivity analyses excluded participants with prevalent cardiovascular disease (stroke, coronary heart disease, and heart failure; self-reported physician's diagnosis at baseline or by standardized adjudication thereafter).^{33,34}

Statistical Analysis

Associations of Late-Life Vascular Risk Factors and Incident Dementia

We used Cox proportional hazards models with the Efron method for ties to estimate cause-specific hazard ratios (HRs) and 95% CIs of incident dementia, and of death without dementia, to assess the independent associations of vascular risk factors and frailty with dementia. The primary inference model included all primary vascular risk factors (hypertension, diabetes, and smoking); frailty; and covariates of age, sex, race, education, alcohol consumption, and total and high-density cholesterol; we also adjusted for APOE ε4 genotype to increase model precision. The proportional hazards assumption was verified by inspecting plots of scaled Schoenfeld residual against transformed survival time. We imputed missing frailty, exposure, and covariate data with multiple imputations with chained equations, with 20 sets of imputations (eMethods 1–2) for all Cox regression models.³⁵ We tested for statistical multiplicative interaction between vascular factors and frailty on the relative risk of dementia in the full sample by including interaction terms (hypertension*frailty, diabetes*frailty, smoking*frailty) in 3 separate Cox regression models adjusted for the primary inference model covariates.

Associations of Late-Life Vascular Risk Factors and Incident Dementia by Frailty Status

After performing tests for interaction, we stratified the full sample and estimated cumulative risks and cause-specific HRs for dementia in robust and in prefrail/frail subgroups. We estimated the cumulative incidence of dementia accounting for the competing risk of death using the cumulative incidence function and Aalen-Johansen estimator adjusted with inverse probability of exposure weights (eMethods 1–2).^{36,37} We estimated cause-specific HRs and 95% CIs of dementia, and of death without dementia, with Cox regression models, which included all

vascular risk factors (hypertension, diabetes, and smoking) and primary inference model covariates except frailty status.

Because there could be reverse causation from existing neurodegeneration or cognitive impairment affecting vascular risk, we also separately estimated HRs after excluding events within ≤ 2 years, ≤ 5 years, and ≤ 8 years of follow-up.³⁸ In exploratory analyses, we tested for heterogeneity of associations in the robust and prefrail/frail samples by fitting interaction terms between hypertension and age at risk factor measurement (<75 years, ≥ 75 years), sex, race, and APOE $\epsilon 4$ genotype on dementia in separate regression models; we also estimated HRs stratified by these 4 variables.

Sensitivity Analyses

In sensitivity analyses, we estimated HRs of dementia after excluding those with prevalent cardiovascular disease and using JNC-7 criteria for hypertension.²⁹ We assessed the effect of missing data by estimating HRs of dementia and for death without dementia only in participants with complete data. We estimated HRs of dementia separately for hypertension, diabetes, and smoking in 3 separate Cox regression models that do not include the other 2 primary vascular risk factors of interest as covariates (e.g., a regression model for hypertension without inclusion of diabetes or smoking as covariates). Because cumulative comorbidity burden can accelerate functional decline, we also estimated HRs of dementia with a primary variable for total vascular risk factor burden (i.e., none present, 1 present, 2 present, and all 3 present).

We used Wald χ^2 tests with a 2-sided $p < 0.05$ to assess statistical significance of interaction terms in the primary analysis; these were specified a priori and therefore not adjusted for multiplicity.³⁹ All analyses were performed in R version 4.2.0 (R Foundation for Statistical Computing, Vienna, Austria, 2021).

Standard Protocol Approvals, Registrations, and Patient Consents

We performed data analysis in August 2024-May 2025. The study was approved by each center's institutional review board. Written informed consent was obtained from all participants. This study conforms to Strengthening the Reporting of Observational Studies in Epidemiology guidelines.

Data Availability

ARIC data are publicly available through visit 6 (biolincc.nhlbi.nih.gov/studies/aric/). Interested researchers may also contact the ARIC Coordinating Center at the University of North Carolina.

Results

Cohort Characteristics

Among participants at baseline (robust, $n = 2,383$; prefrail $n = 2,376$; frail, $n = 334$ with complete data; eTable 1), the mean

age was 75.4 years (SD, 5.1 years; range, 67–89 years), 2,981 (58.5%) were female, and 1,059 (20.8%) were Black (Table 1). Overall, 4,127 (81.0%) had hypertension, 1,611 (31.6%) had diabetes, and 318 (6.2%) were current smokers. Over a median follow-up of 9.0 (25th–75th percentile, 6.0–9.8) years, there were 1,189 (23.3%) incident dementia cases and 1,056 (20.7%) participant deaths. Dementia incidence was higher in prefrail/frail participants ($n = 812$, 30.0%) relative to robust participants ($n = 377$, 15.8%) (Figure 1).

Associations of Late-Life Vascular Risk Factors and Incident Dementia

In the full sample, diabetes (HR = 1.29; 95% CI 1.15–1.44), current smoking (HR = 1.50; 95% CI 1.19–1.89), and prefrail/frail status (HR = 1.59; 95% CI 1.43–1.78) were independently associated with a higher risk of dementia (eTable 2). HRs for elevated BP and hypertension were 0.77 (95% CI 0.59–1.00) and 0.97 (95% CI 0.82–1.16), respectively.

There was an interaction between hypertension ($p = 0.026$) with frailty status, which indicated an inverse association of hypertension with dementia (i.e., lower relative risk) in prefrail/frail participants and a positive association with dementia (i.e., higher relative risk) in robust participants. There was no interaction between diabetes or smoking and frailty status ($p = 0.883$ and $p = 0.367$, respectively), indicative of positive associations between diabetes and smoking with dementia incidence irrespective of frailty status.

Associations of Late-Life Vascular Risk Factors and Incident Dementia by Frailty Status

With stratification of the full sample to compare robust with prefrail/frail participants, adjusted HRs of dementia for elevated BP were 1.03 (95% CI 0.65–1.64) in robust participants vs 0.68 (95% CI 0.49–0.95) in prefrail/frail participants (Table 2). HRs of dementia for hypertension were 1.39 (95% CI 1.00–1.94) in robust participants vs 0.82 (95% CI 0.66–1.01) in prefrail/frail participants (Table 2). By severity of hypertension, the HR for stage 1 hypertension in robust participants was 1.66 (95% CI 1.07–2.56) vs 0.80 (95% CI 0.56–1.12) in prefrail/frail participants; and the HR for stage 2 hypertension was 1.36 (95% CI 0.97–1.90) vs 0.82 (95% CI 0.66–1.01), respectively. HRs of dementia for diabetes and for current smoking in robust participants were similar to those estimated in prefrail/frail participants (Table 2).

Hypertension, diabetes, and smoking were all associated with higher risk of non-dementia mortality in both frailty strata (Table 2). For example, comparing robust with prefrail/frail participants, HRs of death for hypertension were 1.18 (95% CI 0.86–1.63) vs 1.19 (95% CI 0.92–1.53), respectively.

HRs of dementia for hypertension, diabetes, and smoking stratified by ordinal frailty category (robust, prefrail, and frail) were in the same direction as the primary analysis but with wider CIs (eTable 3). Excluding events within ≤ 2 years, ≤ 5 years,

Table 1 Cohort Characteristics, ARIC-NCS (2011–2022)

	Baseline (2011–2013)			Last status during follow-up (2011–2022)								
				Alive, no dementia			Dementia			Deceased, no dementia		
	Total	Robust	Prefrail/ frail	Total	Robust	Prefrail/ frail	Total	Robust	Prefrail/ frail	Total	Robust	Prefrail/ frail
N	5,093	2,383	2,710	2,848	1,625	1,223	1,189	377	812	1,056	381	675
Age, y^a	75.4 (5.1)	74.3 (4.6)	76.4 (5.3)	73.8 (4.3)	73.4 (4.0)	74.5 (4.5)	77.8 (5.2)	76.7 (4.8)	78.3 (5.3)	77.0 (5.3)	75.8 (5.0)	77.6 (5.4)
Female, N (%)	2,981 (58.5)	1,319 (55.4)	1,662 (61.3)	1786 (62.7)	959 (59.0)	827 (67.6)	702 (59.0)	200 (53.1)	502 (61.8)	493 (46.7)	160 (42.0)	333 (49.3)
Black, N (%)	1,059 (20.8)	445 (18.7)	614 (22.7)	557 (19.6)	292 (18.0)	265 (21.7)	291 (24.5)	89 (23.6)	202 (24.9)	211 (20.0)	64 (16.8)	147 (21.8)
Education level, N (%)												
Lower than completed HS	640 (12.6)	222 (9.3)	418 (15.4)	244 (8.6)	108 (6.6)	136 (11.1)	232 (19.5)	67 (17.8)	165 (20.3)	164 (15.5)	47 (12.3)	117 (17.3)
HS or equivalent	2,165 (42.5)	943 (39.6)	1,222 (45.1)	1,213 (42.6)	646 (39.8)	567 (46.4)	495 (41.6)	141 (37.4)	354 (43.6)	457 (43.3)	156 (40.9)	301 (44.6)
At least some college	2,288 (44.9)	1,218 (51.1)	1,070 (39.5)	1,391 (48.8)	871 (53.6)	520 (42.5)	462 (38.9)	169 (44.8)	293 (36.1)	435 (41.2)	178 (46.7)	257 (38.1)
≥1 APOE ε4 allele, N (%)	1,419 (27.9)	656 (27.5)	763 (28.2)	710 (24.9)	417 (25.7)	293 (24.0)	422 (35.5)	140 (37.1)	282 (34.7)	287 (27.2)	99 (26.0)	188 (27.9)
Hypertension, N (%)												
Normal BP	571 (11.2)	289 (12.1)	282 (10.4)	357 (12.5)	220 (13.5)	137 (11.2)	124 (10.4)	32 (8.5)	92 (11.3)	90 (8.5)	37 (9.7)	53 (7.9)
Elevated BP	395 (7.8)	228 (9.6)	167 (6.2)	263 (9.2)	168 (10.3)	95 (7.8)	66 (5.6)	27 (7.2)	39 (4.8)	66 (6.2)	33 (8.7)	33 (4.9)
Hypertension	4,127 (81.0)	1866 (78.3)	2,261 (83.4)	2,228 (78.2)	1,237 (76.1)	991 (81.0)	999 (84.0)	318 (84.4)	681 (83.9)	900 (85.2)	311 (81.6)	589 (87.3)
BP medication, N (%)	3,409 (66.9)	1,483 (62.2)	1,926 (71.1)	1803 (63.3)	973 (59.9)	830 (67.9)	831 (69.9)	252 (66.8)	579 (71.3)	775 (73.4)	258 (67.7)	517 (76.6)
Diabetes, N (%)	1,611 (31.6)	617 (25.9)	994 (36.7)	771 (27.1)	375 (23.1)	396 (32.4)	438 (36.8)	112 (29.7)	326 (40.1)	402 (38.1)	130 (34.1)	272 (40.3)
Smoking status, N (%)												
Never	2,103 (41.3)	982 (41.2)	1,121 (41.4)	1,230 (43.2)	689 (42.4)	541 (44.2)	517 (43.5)	167 (44.3)	350 (43.1)	356 (33.7)	126 (33.1)	230 (34.1)
Former	2,672 (52.5)	1,267 (53.2)	1,405 (51.8)	1,470 (51.6)	858 (52.8)	612 (50.0)	592 (49.8)	190 (50.4)	402 (49.5)	610 (57.8)	219 (57.5)	391 (57.9)
Current	318 (6.2)	134 (5.6)	184 (6.8)	148 (5.2)	78 (4.8)	70 (5.7)	80 (6.7)	20 (5.3)	60 (7.4)	90 (8.5)	36 (9.4)	54 (8.0)
Alcohol use, N (%)												
Never	1,066 (20.9)	1,366 (57.3)	1,196 (44.1)	552 (19.4)	968 (59.6)	588 (48.1)	318 (26.7)	172 (45.6)	304 (37.4)	196 (18.6)	226 (59.3)	304 (45.0)
Former	1,465 (28.8)	593 (24.9)	872 (32.2)	740 (26.0)	388 (23.9)	352 (28.8)	395 (33.2)	109 (28.9)	286 (35.2)	330 (31.2)	96 (25.2)	234 (34.7)
Current	2,562 (50.3)	424 (17.8)	642 (23.7)	1,556 (54.6)	269 (16.6)	283 (23.1)	476 (40.0)	96 (25.5)	222 (27.3)	530 (50.2)	59 (15.5)	137 (20.3)
HDL-C, mg/dL^a	52.3 (13.9)	52.6 (13.7)	52.0 (14.2)	53.3 (14.0)	53.3 (13.7)	53.3 (14.5)	51.9 (13.6)	52.1 (13.4)	51.8 (13.6)	50.0 (13.8)	50.0 (13.5)	50.0 (14.0)

Continued

Table 1 Cohort Characteristics, ARIC-NCS (2011–2022) (continued)

	Baseline (2011–2013)			Last status during follow-up (2011–2022)								
				Alive, no dementia			Dementia			Deceased, no dementia		
	Total	Robust	Prefrail/ frail	Total	Robust	Prefrail/ frail	Total	Robust	Prefrail/ frail	Total	Robust	Prefrail/ frail
Total cholesterol, mg/dL^a	181.8 (41.7)	183.8 (40.2)	179.9 (42.9)	185.5 (40.2)	185.7 (38.9)	185.1 (41.8)	179.3 (41.8)	183.2 (43.5)	177.5 (40.9)	174.5 (44.4)	176.2 (41.3)	173.5 (46.0)
BMI, kg/m²^b	27.9 (24.9, 31.7)	27.5 (24.8, 30.9)	28.4 (25.0, 32.3)	28.1 (25.1, 31.7)	27.6 (24.9, 30.9)	28.9 (25.4, 32.6)	27.5 (24.6, 31.3)	27.1 (24.9, 30.6)	27.7 (24.4, 31.7)	27.9 (24.8, 32.0)	27.4 (24.4, 31.4)	28.3 (25.0, 32.2)
Follow-up, y^b	9.0 (6.0, 9.8)	9.2 (7.8, 9.9)	8.2 (4.9, 9.7)	9.6 (9.0, 10.1)	9.6 (9.0, 10.1)	9.6 (9.0, 10.0)	5.9 (4.4, 8.2)	6.7 (5.2, 8.7)	5.5 (4.1, 7.9)	5.7 (4.5, 7.9)	6.5 (4.5, 8.6)	5.2 (4.4, 7.5)

Abbreviations: APOE = apolipoprotein; BMI = body mass index; BP = blood pressure; HDL-C = high-density lipid cholesterol.

^a Reported as mean (SD).

^b Reported as median (25th, 75th percentile).

and ≤8 years of follow-up resulted in HRs consistent with the primary analysis but with lower precision (Table 3). Tests for heterogeneity of associations between hypertension with risk of dementia by baseline age, sex, race, or APOE ε4 genotype all resulted in wide and overlapping CIs (Figure 2).

Sensitivity Analyses

Exclusion of participants with baseline cardiovascular disease yielded HRs consistent with the primary analysis (eTable 4). There was statistically significant multiplicative interaction between JNC-7-defined hypertension with frailty on risk of dementia ($p = 0.025$); subsequent HRs stratified by robust vs prefrail/frail status were consistent with the primary analysis (eTable 5). In the complete-case analysis, HRs of dementia and of death without dementia were consistent with the primary analysis (eTable 6). HRs of dementia for each primary vascular risk factor individually that were not simultaneously adjusted for the other 2 were also consistent with the primary analysis (eTable 7). There was also statistically significant interaction between vascular risk factor burden and frailty ($p = 0.033$), with stratified HRs for dementia stronger in robust vs prefrail/frail participants (eTable 8).

Discussion

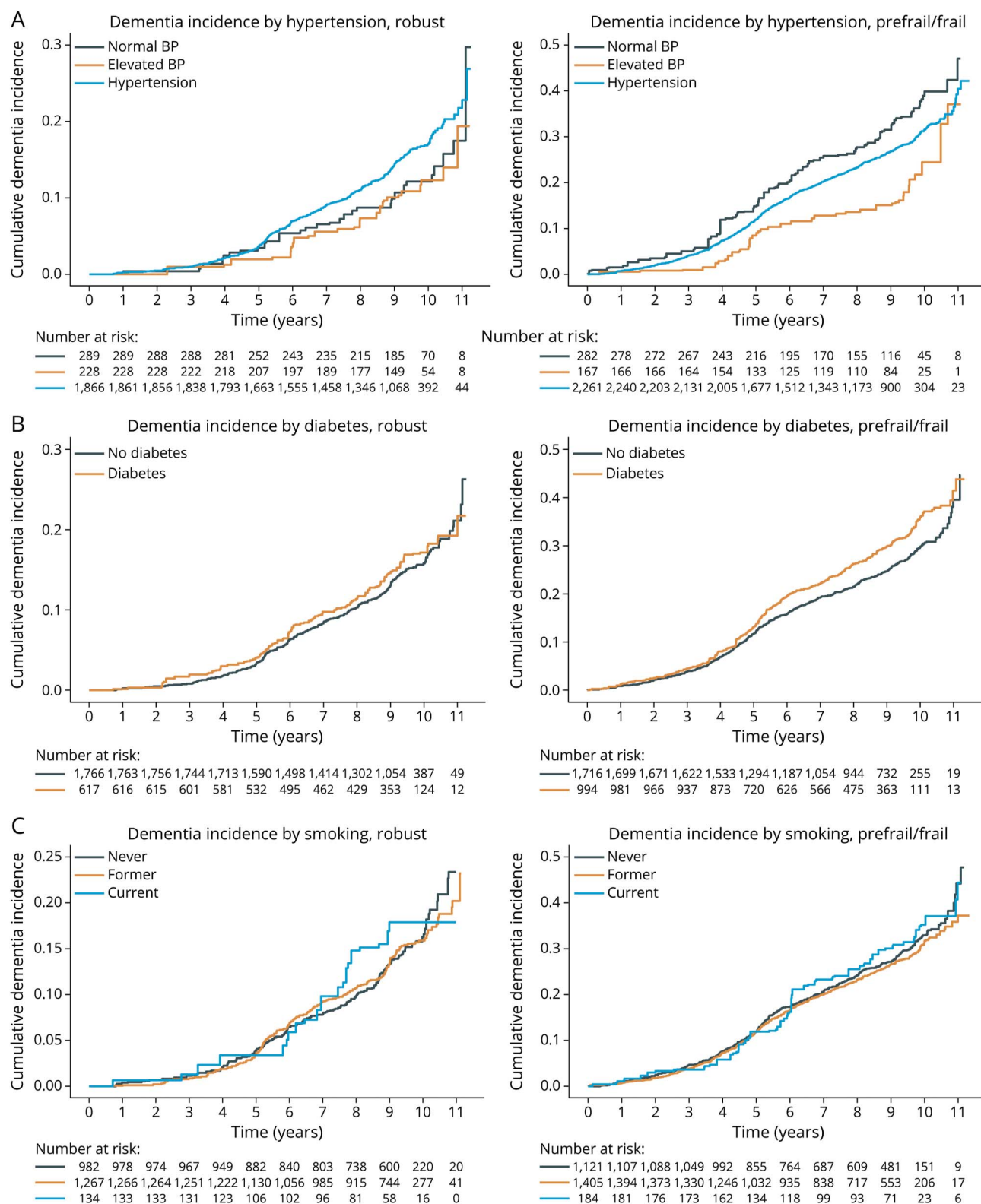
In the community-based ARIC-NCS cohort, we investigated associations of vascular risk factors with incident dementia over 11 years by frailty status in older adults aged 67–89 years at baseline. We found that late-life hypertension significantly interacted with frailty, with a negative relative risk of dementia in prefrail/frail older adults and a positive relative risk in robust adults. However, acknowledging the limitations of wide CIs, we found no differences in associations for late-life diabetes and current smoking by frailty status.

Few studies have prospectively evaluated vascular risk factor associations with dementia incidence by frailty. One study of

older adults aged on average 66 years at baseline—younger on average than in our sample—found no difference in dementia incidence assessed in the Korean National Health Information Database from stage 2 hypertension by frailty over a mean follow-up of 6.6 years.⁴⁰ Post hoc evidence from the SPRINT trial (mean age, 67.9 years), however, suggested aggressive BP control was associated with a lower relative risk of cognitive impairment in robust participants (HR = 0.70; 95% CI 0.59–0.84) and higher relative risk in frail participants (HR = 1.88; 95% CI 1.56–2.27) ($p < 0.05$) over a median follow-up of 4.6 years.⁴¹ Because the adverse effects of hypertension on subsequent dementia risk are presumed to accrue over a longer duration, our study adds new information with extended follow-up (up to 11 years), inclusion of participants older than 80 years, and those with additional comorbidities (e.g., diabetes).

That elevated BP and hypertension compared with normal BP were associated with a lower relative risk of dementia in prefrail/frail participants may have a biologic explanation. The brain in frail participants could be particularly vulnerable to reduced BP, with adequate BP levels needed to prevent hypoperfusion. Prior studies in ARIC and elsewhere indicated late-life hypertension predicts dementia poorly,¹⁰ which is supported by the modest SPRINT-MIND and recent China Rural Hypertension Control Project Phase 3 (CRHCP-3) trial results for probable all-cause dementia.^{42–44} Research in ARIC¹⁸ further demonstrated that the highest relative risk of dementia was in persons with a pattern of midlife hypertension followed by late-life hypotension. It was postulated that those findings were the result of long-accumulating sclerosis of small vessels, which cannot maintain forward blood flow in the brain without adequate BP. After stratification on frailty status, our study suggests this phenomenon could apply specifically to the brain for prefrail/frail, but not robust, individuals. However, other adverse effects (e.g., renal injury) cannot be ruled out. While we found hypertension was associated with a lower relative risk of dementia in prefrail/frail

Figure 1 Cumulative Incidence of Dementia by Vascular Risk Factor Status and Frailty, ARIC-NCS (2011–2022)



The cumulative incidence of 11-year dementia in older adults by hypertension (panel A), diabetes (panel B), and smoking (panel C) and stratified by frailty (robust vs prefrail/frail). Estimates are derived from the cumulative incidence function to account for the competing risk of death and adjusted using stabilized inverse probability of exposure weights. BP = blood pressure.

Table 2 Associations of Late-Life Vascular Risk Factors at Visit 5 (2011–2013) With Incident Dementia by Frailty, ARIC-NCS (2011–2022)

Risk factor	Crude No. of dementia cases/total No. exposed (crude case rate per 1,000 person-years) ^a		HR for dementia (95% CI) ^b		<i>p</i> Value for interaction ^b	HR for death without dementia (95% CI) ^b	
	Robust	Prefrail/frail	Robust	Prefrail/frail		Robust	Prefrail/frail
Hypertension					0.026		
Normal BP	32/289 (12.6)	92/282 (42.3)	1 [Reference]	1 [Reference]		1 [Reference]	1 [Reference]
Elevated BP	27/228 (13.3)	39/167 (29.5)	1.03 (0.65–1.64)	0.68 (0.49–0.95)		1.19 (0.77–1.81)	0.84 (0.57–1.23)
Hypertension	318/1866 (20.2)	681/2,261 (41.2)	1.39 (1.00–1.94)	0.82 (0.66–1.01)		1.18 (0.86–1.63)	1.19 (0.92–1.53)
Diabetes	112/617 (22.1)	326/994 (45.5)	1.28 (1.04–1.57)	1.29 (1.12–1.47)	0.883	1.68 (1.38–2.05)	1.27 (1.09–1.47)
Smoking					0.367		
Never	167/982 (19.8)	350/1,121 (41.6)	1 [Reference]	1 [Reference]		1 [Reference]	1 [Reference]
Former	190/1,267 (17.6)	402/1,405 (39.0)	1.18 (0.97–1.45)	0.96 (0.83–1.12)		1.24 (1.00, 1.54)	1.17 (0.99–1.37)
Current	20/134 (18.9)	60/184 (44.8)	1.59 (1.02–2.48)	1.44 (1.11–1.89)		2.46 (1.69–3.59)	1.59 (1.20–2.10)

Abbreviations: BP = blood pressure; HR = hazard ratio.
^a Crude dementia cases and case rate/1,000 person-years are calculated in a complete case sample (robust, n = 2,383; prefrail/frail, n = 2,710).
^b Cox regression model covariates include hypertension, diabetes, smoking status, age, sex, race, education, apolipoprotein ε4 genotype, alcohol consumption, high-density cholesterol and total cholesterol; HRs and Wald test *p* values for interaction are pooled using Rubin's rules.

participants, it remained associated with higher non-dementia mortality risk.

We cannot rule out the association of dementia with lower BP having noncausal explanations. First, BP may decrease in persons who are in late or asymptomatic stages of dementia.¹¹ Our findings were consistent, however, after excluding events within 2, 5, or 8 years of follow-up, and this reduces the likelihood of reverse causation. Alternatively, dementia after low BP could be attributed to survivor bias (left truncation). Frail participants who are included are survivors of higher mortality rates than robust participants, and higher mortality may have already removed participants with hypertension at higher dementia risk (e.g., more comorbid risk factors).¹⁸ While plausible, this competing risk hypothesis assumes a selective effect strong enough to induce a qualitatively different association between hypertension and dementia. It also does not explain why hypertension in frail participants remained associated with mortality without dementia. Finally, there could be residual confounding from comorbidity; but this also provides little explanation of the mortality findings. These noncausal explanations cannot be ruled out, and replication in studies with more information on the course and characteristics of hypertension in frail and robust individuals is needed to support (or refute) the biologic explanation of our results. Future studies are also needed to investigate whether these results apply to specific dementia subtypes, in particular vascular dementia, beyond all-cause dementia.

Our findings highlight a holistic consideration of overall health context and functional status is likely needed to guide BP management in older adults.^{12,13,45} The benefits of low BP

for reducing cardiovascular-related morbidity and mortality irrespective of frailty status, supported by our mortality findings and by SPRINT trial results in those aged 75 years and older,^{46,47} might need to be weighed against potential risks of low BP on brain health and other potential adverse effects. Our results indicate control of vascular risk factors in late life might support cognition even into the eighth decade of life in robust older adults, meaning selection of optimal BP targets for preserving brain health in older adults could take into consideration frailty status.¹³ This could also have implications for future clinical trials. Future studies that investigate the effect of different BP treatment strategies on brain health in older individuals might benefit from restricted enrollment or subgroup analysis stratified by frailty.

There are several limitations to our study. We did not account for age of vascular risk factor onset, control, or their cumulative effects or trajectories, which could affect our estimates. We did not evaluate effects of hypertension or diabetes treatment strategies on reducing dementia risk; however, we included treatment in our definitions for hypertension and diabetes in accordance with guideline definitions. Although we used a standard definition to define frailty, our primary analysis combined prefrailty and frailty subgroups (no components present vs at least 1) because of sample size considerations. However, results were consistent, albeit with wider CIs, in analyses that used the 3-level categorization. We did not evaluate whether frailty is a potential mediator of associations between any vascular risk factors and dementia. Second-order interaction analyses were likely underpowered. Only White and Black individuals were represented in this analysis, and all third-order interaction analyses (e.g.,

Table 3 Associations of Late-Life Vascular Risk Factors With Incident Dementia by Frailty Status With Different Landmarks, ARIC-NCS (2011–2022)

	No. of dementia cases/total No. exposed (crude case rate per 1,000 person-years) ^a		HR for dementia (95% CI) ^b		<i>p</i> Value for interaction ^b
	Robust	Prefrail/frail	Robust	Prefrail/frail	
2-y landmark					
Hypertension					0.058
Normal BP	31/288 (15.8)	83/273 (51.1)	1 [Reference]	1 [Reference]	
Elevated BP	27/228 (17.1)	37/165 (37.4)	1.06 (0.66–1.68)	0.71 (0.51–0.99)	
Hypertension	308/1855 (25.6)	634/2,201 (52.5)	1.37 (0.98–1.92)	0.84 (0.68–1.05)	
Diabetes	109/614 (28.4)	304/969 (58.4)	1.28 (1.04–1.58)	1.29 (1.12–1.49)	0.863
Smoking					0.459
Never	160/974 (24.6)	322/1,087 (52.0)	1 [Reference]	1 [Reference]	
Former	187/1,264 (22.6)	378/1,376 (50.3)	1.20 (0.98–1.48)	0.99 (0.85–1.16)	
Current	19/133 (23.9)	54/176 (55.2)	1.52 (0.96–2.43)	1.44 (1.09–1.91)	
5-y landmark					
Hypertension					0.015
Normal BP	23/256 (20.7)	62/226 (72.7)	1 [Reference]	1 [Reference]	
Elevated BP	24/213 (26.4)	24/130 (45.8)	1.27 (0.75–2.15)	0.65 (0.43–0.98)	
Hypertension	246/1,655 (37.0)	404/1,663 (67.1)	1.51 (1.01–2.24)	0.74 (0.57–0.97)	
Diabetes	86/525 (41.3)	204/724 (80.2)	1.22 (0.95–1.56)	1.37 (1.15–1.64)	0.443
Smoking					0.814
Never	128/887 (35.1)	213/857 (67.0)	1 [Reference]	1 [Reference]	
Former	149/1,129 (32.4)	239/1,025 (64.0)	1.15 (0.91–1.46)	0.98 (0.81–1.18)	
Current	16/108 (38.2)	38/137 (78.3)	1.58 (0.94–2.66)	1.52 (1.09–2.11)	
8-y landmark					
Hypertension					0.071
Normal BP	12/221 (31.9)	30/164 (110.3)	1 [Reference]	1 [Reference]	
Elevated BP	11/180 (34.8)	14/106 (84.3)	1.24 (0.56–2.75)	0.72 (0.39–1.34)	
Hypertension	106/1,328 (49.4)	152/1,153 (84.9)	1.63 (0.90–2.93)	0.75 (0.51–1.12)	
Diabetes	36/412 (53.6)	74/477 (99.4)	1.19 (0.81–1.75)	1.24 (0.93–1.67)	0.892
Smoking					0.500
Never	64/731 (53.4)	88/620 (90.4)	1 [Reference]	1 [Reference]	
Former	61/915 (40.1)	89/707 (80.0)	0.95 (0.66–1.36)	1.01 (0.74–1.38)	
Current	4/83 (33.6)	19/96 (133.9)	1.09 (0.41–2.95)	1.80 (1.07–3.01)	

Abbreviations: BP = blood pressure; HR = hazard ratio.

^a Crude dementia cases and case rate/1,000 person-years are calculated among complete cases.

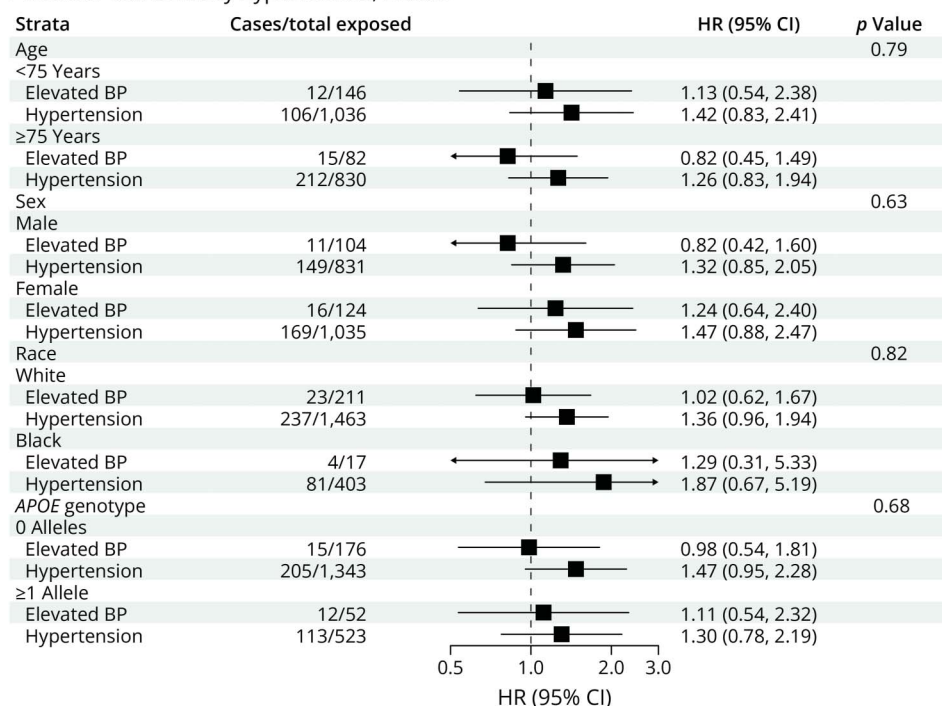
^b Cox regression model covariates include hypertension, diabetes, smoking status, age, sex, race, education level, apolipoprotein ε4 genotype, alcohol consumption, high-density cholesterol, and total cholesterol; HRs and Wald test *p* values for interaction are pooled using Rubin rules.

hypertension*frailty*race) are underpowered and considered exploratory. This is important to evaluate known disparities because Black individuals have a higher documented prevalence of vascular risk factors and of dementia in the

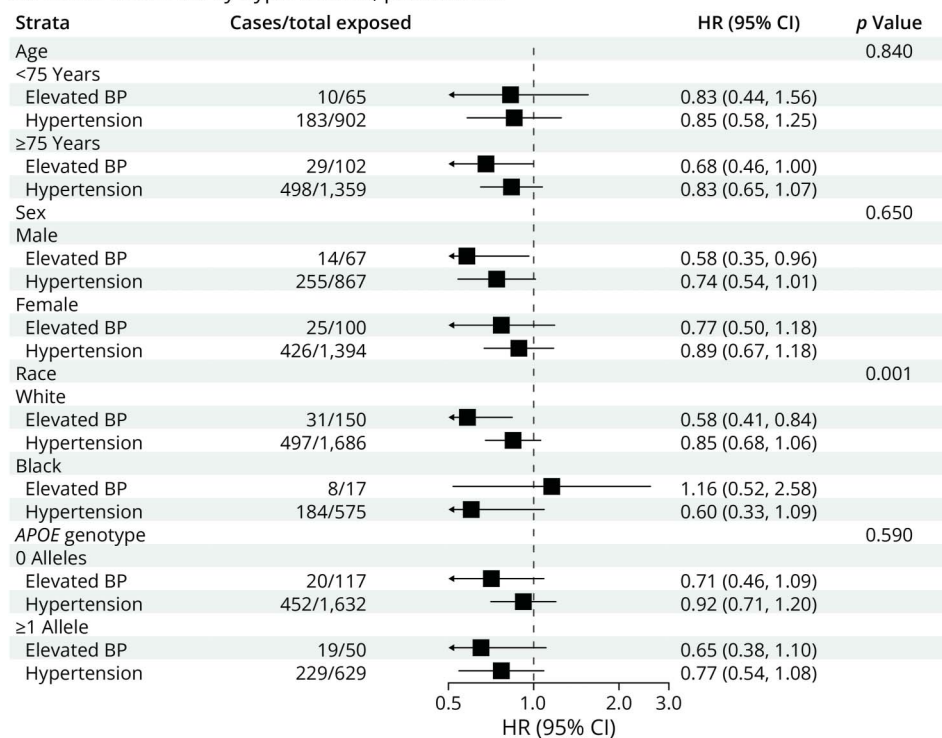
United States. Future studies with greater power are needed to investigate associations in different racial and ethnic populations with varying vascular risk profiles in older age. Finally, only all-cause dementia is available in ARIC, and we could not

Figure 2 Stratified Associations of Late-Life Hypertension and Frailty Status With Incident Dementia, ARIC-NCS (2011–2022)

A. HR for dementia by hypertension, robust



B. HR for dementia by hypertension, prefrail/frail



Stratified associations of late-life hypertension with incident dementia in robust (panel A) vs prefrail/frail (panel B) older adults. HRs are estimated from Cox regression models adjusted for age, sex, race, education level, *APOE* ϵ 4 genotype, alcohol consumption, and total and high-density cholesterol. Estimates stratified by age at risk factor measurement (<75 years or ≥75 years), sex (male or female), self-identified race (White or Black), and *APOE* ϵ 4 genotype (0 alleles or ≥1 allele). *APOE* = apolipoprotein; BP = blood pressure; HR = hazard ratio.

examine associations of vascular factors with subtypes of dementia. It is likely that the strongest associations would be seen between vascular risk factors—hypertension in particular—and vascular dementia. In ARIC, neuroimaging is only available in a subset of participants. Vascular dementia

and Alzheimer disease are also difficult to distinguish clinically, or even with neuroimaging and autopsy, given the substantially high prevalence of brain copathologies in later life that contribute to the heterogeneous clinical expression of dementia.^{48,49}

In the ARIC-NCS cohort, hypertension significantly interacted with frailty, whereby hypertension was associated with elevated dementia risk over 11 years in robust older adults and lower dementia risk in prefrail/frail adults. BP management in late life to maintain brain health might need to consider frailty status in the context of patient-centered care. Future studies are needed to investigate effects of aggressive treatment strategies in late life on lowering risk of dementia by frailty status.

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Author Contributions

J.R. Smith: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. A.R. Sharrett: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. J.R. Pike: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. R.F. Gottesman: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. D.S. Knopman: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. B.G. Windham: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. P.L. Lutsey: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. P. Palta: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J. Coresh: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. K. Bandeen-Roche: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. A.L. Gross: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. J.A. Deal: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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Disclosure

D.S. Knopman serves on a data safety monitoring board for the Dominantly Inherited Alzheimer Network Treatment Unit study. He served on a data safety monitoring board for a tau therapeutic for Biogen (until 2021) but received no personal compensation. He is an investigator in clinical trials sponsored by Biogen, Lilly Pharmaceuticals, and the University of Southern California. He has served as a consultant for Roche, Samus Therapeutics, Magellan Health, BioVie, and Alzeca Biosciences but receives no personal compensation. He attended an Eisai advisory board meeting for lecanemab on December 2, 2022, but received no compensation. He receives funding from the NIH. No other disclosures are reported. Go to [Neurology.org/N](https://www.neurology.org/N) for full disclosures.

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